
When Coverage Decides the Ranking: A Pre-Registered Audit of Completion-Conditioned Computer-Use Agent Evaluation

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Abstract

1 Computer-use agents are compared by a completion-conditioned score: an
2 episode must terminate, and a rubric judge then assigns a number. We ask whether
3 that number and a state-tracking measure select the same agent, on a universe and
4 analysis plan frozen before any outcome existed. Three agents each ran 57 pre-
5 registered legs under one instrument. Coverage was the first result rather than a
6 nuisance: valid counterfactual pairs, which require both legs to reach a canonical
7 DONE, numbered 9 for GPT-5.5, 8 for Qwen3.8-Flash, and 1 for Claude Opus 4.6,
8 so only two agents met the pre-registered threshold $n_{\min} = 3$ and the confirma-
9 tory selection test was, by its own pre-registered rule, not evaluated. Binary state
10 tracking was 0 on *every* valid pair of every agent, leaving score-versus-tracking
11 calibration degenerate rather than weak. In the remaining exploratory compari-
12 son the two measurement layers did not agree: on a four-task common support
13 the score favoured Flash by +46.5 points (95% CI [21.0, 72.0]), while graded
14 tracking favoured GPT-5.5 by -0.042 (95% CI $[-0.125, 0.000]$) — an aggregate
15 reversal that is strict on one of four tasks, tied on the other three, and erased by
16 removing that one task. We therefore report measurement instability under this
17 instrument, not an agent ranking: the observed ordering depends on which layer
18 and which support is used.

19 1 Introduction

20 Leaderboards for computer-use agents (CUAs) report a completion score: the episode ended and
21 a rubric assigned a number [Jang et al., 2026, Xie et al., 2024]. Selecting an agent from such a
22 leaderboard presumes that the score orders agents the way the property one cares about would order
23 them. Prior work in this line showed that, for individual episodes, task completion, counterfactual
24 state tracking, and score sensitivity can come apart [Shao et al., 2026, Turk, 2026]. That leaves
25 the decision-relevant question open: when the measurement layer changes, does the *choice* of agent
26 change?

27 This paper answers that question on a sealed universe with an analysis plan fixed in advance, and
28 reports what actually happened, which is not a clean disagreement and not a clean agreement. Two
29 findings dominate and both are about the instrument rather than the agents.

30 First, **coverage decided who could be compared at all**. The unit of a counterfactual comparison
31 is a *pair* of episodes, base and intervened, and a pair enters the analysis only if both legs terminate
32 canonically. Termination rates differed so sharply across agents that one frontier agent contributed
33 a single usable pair out of a possible 19, which under the pre-registered threshold removed it from
34 ranking and reduced the roster below the minimum required for the confirmatory test.

35 Second, **the two layers produced different aggregate preferences but at the edge of one layer’s**
 36 **resolution.** Among the two agents with enough support, the conventional score strongly preferred
 37 one agent while graded tracking preferred the other; that reversal, however, is carried entirely by one
 38 task out of four and the tracking measure is tied on the rest.

39 **Contributions.** (1) A pre-registered, coverage-first audit in which the valid-pair count is reported
 40 as a finding rather than a denominator. (2) An instrument whose termination channel is *fail-closed*
 41 and *independent of the judge*, so that “scored well without closing the episode” is an observation
 42 rather than a bookkeeping artefact. (3) An exploratory two-layer comparison reported with its own
 43 fragility: the strict/tied decomposition, the single task that carries the reversal, the resolution limit
 44 of the tracking measure, and the flip between per-agent and common support. The claim is method-
 45 ological and scoped to this instrument; we do not offer a ranking and we do not estimate a population
 46 rate.

47 2 Setup

48 **Frozen universe and plan.** Tasks, agents, interventions, seed, step budget, and the analysis rules
 49 were written and committed before execution. The inclusion threshold ($n_{\min} = 3$ valid pairs to be
 50 ranked), the rule that fewer than three ranked agents means the confirmatory selection test is *not*
 51 *evaluated*, the prohibition on optional stopping, and the form in which a two-agent comparison may
 52 appear were all fixed in advance of any outcome. We note this because every headline number below
 53 is a null or a partial result, and the value of such results depends entirely on the plan predating them.

54 **Pairs and interventions.** For task t , E_t^0 is the seeded guest state and $E_t^1 = I_t(E_t^0)$ applies a
 55 locked patch to a pre-registered determining set D_t . Instruction, interface, screenshot size, and
 56 judge are held fixed; guest probes confirm that gold moved. Each task contributes legs $G0$ (base)
 57 and $G1$ (intervened); a third variant is executed but excluded from pairing by pre-registration. A pair
 58 is *valid* iff both $G0$ and $G1$ terminate canonically, and $\mathcal{A}_i$ denotes agent i ’s valid pairs. Execution
 59 failures are never recoded as tracking failures.

60 **Two measurement layers.** The conventional layer is the base-leg rubric $S^0 \in \{0, \dots, 100\}$, i.e.
 61 the number a leaderboard would report on the unmodified world; base and intervened scores are
 62 deliberately not averaged. The tracking layer is a graded state-tracking score STS, computed by
 63 extracting reported values $\hat{D}$ from the final answer and matching them against guest gold under per-
 64 component typed rules, with no judge in the loop. Binary track $Y = 1$ requires every positive-weight
 65 component to match on *both* legs. The extraction code was frozen at a recorded commit *before* any
 66 aggregate STS was computed, and was not revised afterwards.

67 **Termination is a separate channel, and it fails closed.** Completion is read only from the last
 68 recorded trajectory action. The classifier does not consult the judge’s output when deciding termina-
 69 tion, so episode closure and task score are derived from disjoint inputs; and it is fail-closed, meaning
 70 a missing, malformed, or unreadable trajectory can only be *excluded* from $\mathcal{A}$, never converted into
 71 an apparent completion. Both properties matter for how the numbers may be read: the reported $|\mathcal{A}_i|$
 72 are **lower bounds**, and a high rubric score on a non-terminating episode cannot be an artefact of one
 73 channel leaking into the other. An earlier defect in this project, in which a *truthy done* field was also
 74 set by crash and failure rows, was remediated before the first leg reported here, so no number below
 75 was reclassified after the fact.

76 **Substrate disclosure.** All three lanes ran the same generic XML tool protocol over one chat-
 77 completions endpoint. This is *not* the vendors’ native computer-use instruments: the GPT-5.5 and
 78 Claude lanes were executed on the shared protocol after native tool-ownership validation failed or
 79 was unavailable on the run host. Comparability is therefore within this roster and instrument only,
 80 and is not comparability against natively instrumented results for the same model names.

81 3 Coverage is the first result

82 Each agent ran the full pre-registered schedule of 57 legs; no lane was stopped early or extended.
 83 Table 1 gives the outcome.

Table 1: Coverage under the frozen instrument. A valid pair requires both $G0$ and $G1$ to terminate canonically. $|\mathcal{A}|$ values are lower bounds because the terminal classifier is fail-closed. Claude’s mean S^0 is computed on a single pair.

Agent	DONE/ 57	rate	$ \mathcal{A} $	mean S^0 on $\mathcal{A}$	mean STS on $\mathcal{A}$
GPT-5.5	32	0.561	9	69.3	0.130
Qwen3.8-Flash	29	0.509	8	95.8	0.229
Claude Opus 4.6	4	0.070	1	100 ($n=1$)	0

84 Claude terminated on four legs, and those four fall on three distinct tasks, so exactly one task has
 85 both legs terminating: $|\mathcal{A}| = 1$ out of 19 possible pairs. Under the pre-registered threshold $n_{\min} = 3$
 86 it is reported but not ranked. With one further agent excluded before execution for instrument and
 87 transport mismatch, the ranked roster is two agents, which is fewer than three, and therefore **the**
 88 **confirmatory selection test is not evaluated** — a stated outcome of the plan rather than a failed
 89 experiment.

90 Two readings must be resisted here. Claude’s mean S^0 on $\mathcal{A}$ is the highest in the table, but it
 91 is a single pair on the one task the agent solved and closed; it is a coverage artefact, exactly the
 92 selection pressure the plan anticipated, and not evidence that the agent scores best. Conversely the
 93 non-terminating legs are not tracking failures: they are absent observations, and recoding them as
 94 $Y = 0$ is forbidden by the plan.

95 4 Layer A: calibration is degenerate, not weak

96 The intended calibration analysis asks how well S^0 predicts $P(Y = 1)$ on valid pairs. It cannot be
 97 estimated, for a stronger reason than small n : $Y = 0$ **on all 18 valid pairs across all three agents**.
 98 No pair matched every positive-weight component of D on both legs. There is no positive class, so
 99 S^0 cannot separate $Y = 1$ from $Y = 0$, and any statement of the form “score is (mis)calibrated to
 100 tracking” is unavailable on this sample.

101 The graded measure is correspondingly near its floor: mean STS on $\mathcal{A}$ is 0.130, 0.229, and 0
 102 (Table 1). This is the context in which the next section’s differences must be read. A gap between
 103 STS = 0.250 and STS = 0.208 is a gap between two small residuals, not between a reliable agent
 104 and an unreliable one.

105 5 Exploratory two-layer comparison

106 Because only two agents were rankable, the comparison below is reported in the single
 107 form the plan permits: signed differences on the common support, with a paired boot-
 108 strap, labelled **exploratory**, and not presented as the confirmatory test. The common sup-
 109 port is $\mathcal{A}^\cap = \{\text{counterfactual-f010, preference_inference-f014, retrieval-f002,}$
 110 $\text{retrieval-f009}\}$, $|\mathcal{A}^\cap| = 4$.

111 Aggregate means on $\mathcal{A}^\cap$ point in opposite directions: S^0 is 49.5 for GPT-5.5 versus 96.0 for Flash,
 112 so $\arg \max S^0$ is Flash; STS is 0.250 versus 0.208, so $\arg \max \text{STS}$ is GPT-5.5. A paired bootstrap
 113 over tasks (seed 20260904, $B = 5000$) gives $\Delta S^0 = +46.5$ with 95% CI [21.0, 72.0] and $\Delta \text{STS} =$
 114 -0.042 with 95% CI $[-0.125, 0.000]$, differences taken as Flash minus GPT-5.5.

115 **The reversal is strict on one task and tied on three.** Table 2 gives the per-task values, and they
 116 are less dramatic than the aggregate suggests.

117 On three of four tasks ΔSTS is exactly zero: the tracking layer is *indifferent* between the two
 118 agents, not opposed to the score. Only *retrieval-f009* is a strict reversal. The aggregate ΔSTS
 119 is therefore driven entirely by that one task, and removing it makes the STS comparison a tie. A
 120 leave-one-pair-out check reports exactly this and nothing more: it is a sensitivity analysis confirming
 121 the same single-task concentration, not a second, independent piece of evidence.

122 **The tracking layer is coarse at this scale.** Across these pairs STS takes only the values
 123 $\{0.000, 0.333, 0.500\}$, because each task contributes a handful of typed components. The hon-

Table 2: Per-task values on the common support, Δ as Flash minus GPT-5.5. STS strictly prefers GPT-5.5 on one task and is exactly tied on three.

Task	S^0 GPT	S^0 Flash	STS GPT	STS Flash	ΔS^0	Δ STS
counterfactual-f010	83	100	0.000	0.000	+17	0.000
preference.inference-f014	0	84	0.000	0.000	+84	0.000
retrieval-f002	75	100	0.500	0.500	+25	0.000
retrieval-f009	40	100	0.500	0.333	+60	-0.167

est reading is that the reversal sits at the resolution limit of the measure: with this few components per task, STS has little room to express a small disagreement, and a reader is entitled to ask whether the layer is disagreeing or merely failing to discriminate. We raise the objection rather than wait for it, and we do not claim to settle it on four tasks.

The common support is not a neutral slice. Restricting to $\mathcal{A}^\cap$ moves GPT-5.5’s mean S^0 from 69.3 on its own nine pairs down to 49.5, while Flash’s is essentially unchanged ($95.8 \rightarrow 96.0$). The shared tasks are thus adversarial to GPT-5.5 on the score layer. The direction flips for tracking: on each agent’s own $\mathcal{A}$, Flash is higher on *both* layers (STS 0.229 versus 0.130), whereas on $\mathcal{A}^\cap$ GPT-5.5 is higher on STS. Both denominators are reported and neither was chosen after inspecting which looked better; the discrepancy between them is itself the result, namely that the selection question is sensitive to which tasks each agent happened to finish.

6 Completion-conditioned observability

Conditioning on termination changes which trajectories the downstream measurement can see at all, and the direction of that effect is worth stating precisely because the intuitive guess is wrong. On average, excluded cells score *lower* than included ones: mean S off $\mathcal{A}$ is 46.4, 44.9, and 26.1 for GPT-5.5, Flash, and Claude, against 69.3, 95.8, and 100 on $\mathcal{A}$. The completion filter is therefore conservative for the calibration question rather than flattering.

What survives that averaging is a small set of sharp cases: 2, 2, and 1 excluded cells scored ≥ 90 with no canonical termination. Because the terminal channel is independent of the judge (Section 2), these are genuine observations of substantial rubric-visible progress without episode closure, not scoring accidents. They are descriptive: none enters $\arg \max$, and none is converted into a pair. Their significance is that a completion-conditioned protocol cannot express them, so an agent whose failure mode is non-closure is measured mostly by its absence.

Instrument asymmetry. One further caveat bounds the coverage numbers. A parser canonicalisation that rescues near-miss tool syntax was enumerated from one agent’s dialect, so it is more forgiving of that agent. Re-deriving terminal outcomes with those rescues treated as rejections moves Flash from $|\mathcal{A}| = 8$ to 7; the other lanes are unaffected because the canonicalisation does not fire on their output. The pre-registered dual analysis is reported for this reason rather than assumed away.

7 Discussion

The result of this audit is a measurement problem, not a ranking. Under this frozen instrument, on the four-task common support, with two rankable agents, and with binary tracking identically zero throughout, CUA reliability is not adequately characterised by a single completion-conditioned score: the ordering that measurement produces depends on which layer is used and on which tasks each agent finished. We state the scope inside the claim deliberately. We have shown instability of this instrument on this universe; we have not shown that CUA benchmarks in general are invalid, and the data here could not support that.

Both quantities compared here are proxies. S^0 is a rubric judge’s number and STS is a typed match against guest gold through an extractor; neither is independently established ground truth, so a disagreement between them licenses no inference about which is correct. This is why we decline

164 both directions of the tempting conclusion: the evidence does not say one agent tracks state better,
165 only that the proxies do not order the pair invariantly under the observed support.

166 The constructive implication is for the design of the next measurement rather than for any leader-
167 board. Three properties of this instrument turned out to carry the analysis: termination read from a
168 fail-closed channel, scoring read from a channel that never consults it, and an extraction step frozen
169 before aggregates were computed. Those made the null results interpretable. What they could not
170 supply is a validated target: with $Y = 0$ everywhere, the tracking layer produced an ordering but no
171 positive instance against which to check that the ordering means what its name says. A reliability
172 measure for agents that frequently do not close episodes therefore needs two things this study did not
173 have: a target defined without reference to episode completion, and positive and negative controls
174 that can separate genuine reliability signal from artefacts of support and protocol observability.

175 **Broader impacts.** Making coverage and layer dependence visible discourages reading a single
176 completion score as a certificate of state tracking. The same paired design could be turned on
177 proprietary graders; we intend it as a diagnostic of measurement, not as an exploit.

178 8 Related work

179 OSWorld [Xie et al., 2024], WebArena [Zhou et al., 2024], and VisualWebArena [Koh et al., 2024]
180 score agents with completion rubrics; MyPCBench [Jang et al., 2026] provides the seeded personal
181 desktop that our interventions reuse. OSWorld and WebArena are Apache-2.0, VisualWebArena is
182 MIT, and MyPCBench is MIT with an OSWorld-derived harness under Apache-2.0; we omit URLs
183 to preserve anonymity. None pairs E^0 with $E^1 = I(E^0)$ to separate state tracking from the score,
184 and none reports valid-pair coverage as a quantity that can decide whether a comparison exists.
185 Shao et al. [2026] audit whether the intended capability remains necessary for a score; we ask
186 instead whether two measurement layers select the same agent. Turk [2026] show that coverage-
187 style scores can hide update failures under controlled mutations; the completion-conditioning we
188 report is the same hazard displaced into the eligibility rule rather than the rubric. Bellibatlu et al.
189 [2026] measure judge instability under paraphrase; our instability is with respect to the measurement
190 layer and the analysis support, with the judge held fixed. The paired design follows interventionist
191 language [Pearl, 2009]; we identify no model internals.

192 9 Limitations

193 Three agents, one environment, one instrument, 57 legs each, and 18 valid pairs in total. Only two
194 agents were rankable and the confirmatory selection test was not evaluated, so the two-layer com-
195 parison is exploratory throughout. Binary tracking is zero everywhere, which makes the calibration
196 analysis degenerate and leaves STS unvalidated against any positive instance. The common sup-
197 port has four tasks; the aggregate STS reversal rests on one of them and disappears without it; and
198 STS takes only three distinct values on these pairs. Reported $|\mathcal{A}|$ are lower bounds by construction,
199 and the near-miss asymmetry moves one lane’s count by one pair. The lanes ran a shared generic
200 tool protocol rather than vendor-native computer-use instruments, so these are results for the named
201 models *on this substrate*. Per-step guest state was not retained, so no analysis here can condition on
202 in-episode environment state.

203 10 Conclusion

204 On a universe and analysis plan frozen in advance, the agent that a completion-conditioned score
205 selects is not the agent a graded state-tracking measure selects, but the disagreement is exploratory,
206 concentrated in a single task, and measured at the resolution limit of the tracking layer — and it
207 sits on top of a coverage asymmetry severe enough that one frontier agent yielded a single usable
208 pair and the confirmatory test could not be run at all. The finding we are willing to defend is the
209 instability itself: under this instrument, which agent looks better depends on the measurement layer
210 and the analysis support. Validating a reliability target that does not require episode completion,
211 against explicit controls, is the prerequisite for turning that instability into a measurement.

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Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: The two signed differences on the common support are reported with 95% intervals from a paired bootstrap resampling the four tasks (seed 20260904, $B = 5000$), stated in the abstract and Section 5. The variability captured is resampling over tasks within this frozen universe, not a deployment population, and Section 9 says so. No null-hypothesis test is claimed. Because an interval this wide on four tasks is weak evidence on its own, Section 5 also reports the per-task decomposition and the leave-one-pair-out check, and labels the leave-one-out result as a sensitivity analysis of the same single-task concentration rather than independent confirmation. The tracking interval closes at 0.000, which is reported rather than rounded away.

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Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: No model is trained or fine-tuned. Agent inference ran entirely through one hosted chat-completions endpoint, and the desktop environment ran as a virtual-machine guest on a shared internal cluster; Section 2 discloses that all three lanes used the same generic tool protocol rather than the vendors’ native computer-use instruments, which is the compute-side fact that bounds comparability. The reported execution is 3 agents $\times$ 57 legs = 171 legs. Per-leg wall-clock, host GPU type, and cluster identifiers were not instrumented for this audit and are not claimed. The project consumed more inference than appears here: an additional smaller-model lane was executed and then excluded before analysis for instrument and transport mismatch, and one lane has a pre-patch corpus that was excluded rather than merged.

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